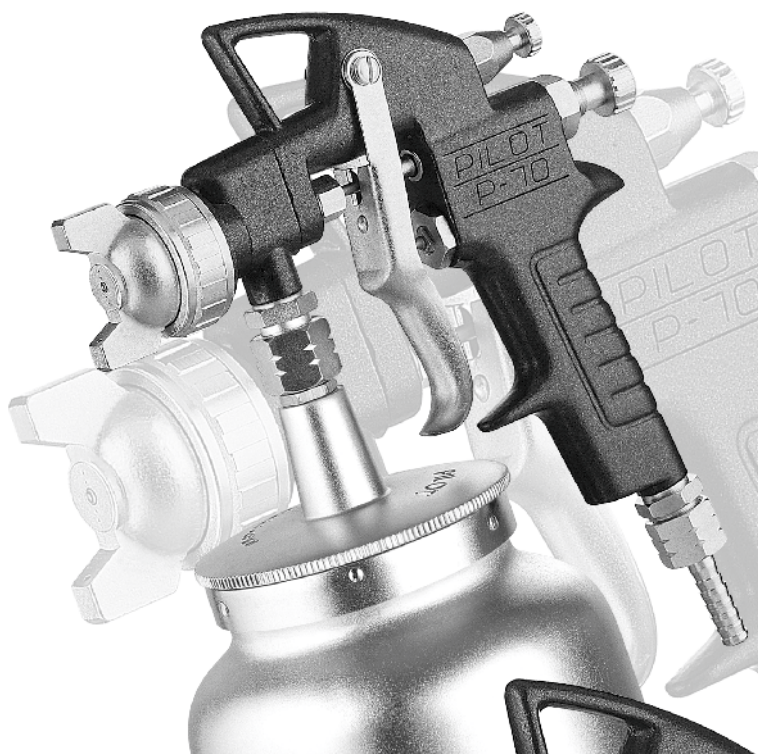




Type  
**P-70**



**PILOT SPRAY GUN**  
Service Sheet & List of Spare Parts

# Dear customer,

Now as you've brought home one of your favourite Pilot spray guns. We'd like to say a special 'thank you'. These precision-engineered tools have been designed to excel, making every spray job the work of a professional.

As you know Pilot stands for quality, reliability and safety, earned by 6 decades of experience and overwhelming customer satisfaction. We are certain we have met all safety requirements. Our products bear the identification plate in conformance to the  ruling.

To facilitate easy understanding of the working procedures please refer to the detailed instruction manual.

## Description

Type - P-70 spray guns are easier to use than most spray guns. These lightweight suction feed spray guns are suitable for production work on articles of medium or large sizes. They are ideal for spray-painting motorcar bodies, refrigerators, machinery, etc. Perfect balance, comfortable handle and trigger action makes them easy and non-tiring to operate. A control knob on the back can easily control round to fan spray of pattern.

These suction feed spray guns are supplied with a 1 lit. suction cup.

**Important:** These guns are not designed to be used with highly corrosive or highly abrasive coating materials. Violation may increase the need for thorough cleaning and /or the necessity for replacement of parts. If there is any doubt regarding the suitability of a specific material you may seek advice and /or submit a sample for test.

## Safety Warnings

### Fire and Explosion

- Solvents and coating materials can be highly flammable or combustible, especially when sprayed.
- Workstations must be provided with adequate ventilation / exhaust to prevent the build up of flammable vapors.
- Smoking and naked flames must not be allowed in the spraying or mixing areas.
- Fire extinguishing equipment must be provided in the spraying and mixing areas.
- Users must comply with all local and national codes of practice and insurance company requirements governing ventilation, fire precautions, operation, maintenance and housekeeping of work stations.

**Halogenated hydrocarbon solvents** for example 1,1,1 Trichloroethane and Methylene Chloride can chemically react with aluminum and galvanized or zinc coated parts and cause an explosion hazard. Read the label and data sheet of the material you intend to spray.

**Do not use solvents or coating materials containing halogenated hydrocarbons with this equipment.**

**Static electricity** is generated by fluid moving through pipes and hoses. A static spark, Capable of igniting certain solvents and coating materials could be produced by high fluid flow rates. To prevent the risk of fire or explosion, earth continuity to the spray equipment and object being sprayed should be maintained.

## Personal protective equipment

**Toxic Vapours** when sprayed, certain material may be poisonous, create irritation or otherwise be harmful to health. Always read carefully all labels and safety / performance data for the material being sprayed and follow any recommendation.

### **If in doubt, consult the material supplier.**

- The use of respiratory protective equipment is recommended at all times when spraying. The type of respiratory protective equipment used must be compatible with the material being sprayed and the level of concentration.
- Always wear eye protection when spraying or cleaning the equipment.
- Gloves must be worn for spraying or cleaning the equipment when certain coating materials and solvents are used.

## Training

Personnel should be given adequate training in the safe use and maintenance of this equipment. Training courses on all aspects of the equipment are available. For details contract your local representative. The instructions and safety precautions contained in this literature and the literature supplied with the coating material should be read and understood before the equipment is used.

## Misuse

- All spray guns project particles at high velocity and must never be aimed at any part of the body.
- Never exceed the recommended safe working pressures for any of the equipment used.  
The fitting of non-recommended or non-original accessories or spare parts may create hazardous conditions.
- Before dismantling the equipment for cleaning or maintenance, all pressures, air and material, must be isolated and released.

The disposal of non-metallic materials must be carried out in an approved manner. Burning may generate toxic fumes. The removal of waste solvents and coating materials should be carried out by an authorized local waste disposal service.

The materials used in the construction of this equipment are (bearing in mind the warning on Halogenated Hydrocarbons) solvent resistant enabling the equipment to be cleaned using gun washing machines. However, this equipment must not be left inside the gun washing machine for prolonged periods of time after the automatic cleaning cycle has been completed. The solvents used in the gun washing machine should be regularly checked to ensure that the equipment is not flushed through with contaminated material. Follow the recommendations of the machine manufacturer.

## Noise Levels

The continuous A-weighted sound pressure level of this spray gun may exceed 85 dB (A) depending on the air cap/nozzle set-up being used. Sound levels are measured using an impulse sound level meter and analyzer, when the gun is being used in a normal spraying application. Details of actual noise levels produced by the various air cap/nozzle set-ups are available on request.

## Installation

**IMPORTANT:** To ensure that this equipment reaches you in first class condition, protective coatings, rust inhibitors, etc. have been used. Flush all equipment through with a suitable solvent before use to remove these agents from the material passages.

- Attach the air supply line to the air intake connector (24). An air transformer installed as close as possible to the gun will provide filtered and regulated air.

### **Recommended air hose sizes:**

- Attached air hose connector from a filtered regulated air supply.  
Recommended air supply hose size up to 10 meters long (1/4") bore.

Attach the suction feed cup to the material inlet and ensure that the sealing washer is compressed.

## **Operation**

Mix, prepare and strain the coating material to be sprayed according to the paint manufacturers instructions. Use a lint free mesh to strain the material.

1. Fill the suction feed cup with the material. Do not overfill. Ensure that the vent hole in the cup lid is clear.
2. Adjust the atomizing air pressure to 2.5 -5.0 kgs/sq.cm. (14 to 72 p.s.i.).
3. It requires air about (7 - 10 c.f.m.) 200 - 283 liter per minute depending on the liquid to be sprayed.
4. Close the needle adjusting screw by turning clockwise.
5. Turn on the air at the source of supply and spray test area by turning the needle adjusting screw counter-clockwise until a full coat is obtained.

If the finish is too sandy and dry, increase the flow of the coating material supply by turning the needle adjusting screw counter-clockwise.

Increasing or decreasing the atomizing air pressure can also correct both the above. The most efficient atomizing air pressure is the lowest possible air pressure that will give the desired effect.

The gun should be held perpendicular to the spray surface at all times. Do not arc the gun as this produces an uneven coat of paint. The recommended spraying distance is 230mm (9").

## **Preventive Maintenance**

### **Cleaning**

Turn off air supply and release pressure.

Empty surplus coating material from cup and clean.

Remove air cap and clean by immersing in solvent, brush or wipe clean. If any holes in the air cap are blocked use a toothpick or broom straw to remove the obstruction. Never use a steel wire or hard implement which will damage the air cap and result in a distorted pattern.

### **Lubrication**

Lubricate all moving parts daily with a few drops of light oil and occasionally place a light coating of Vaseline on the needle spring. The packing may need to be oiled occasionally to keep them pliable.

## **Replacement of Parts**

### **Needle Valve (18)**

1. Remove needle adjusting screw (20), needle spring (19) and needle valve (18). Replace any worn or damaged parts ensuring that the needle valve seats correctly in the nozzle then adjust packing with needle adjusting screw (20) so that the movement of the needle is not restricted.

### **Packing Bush (5)**

Remove needle valve as 1 above.

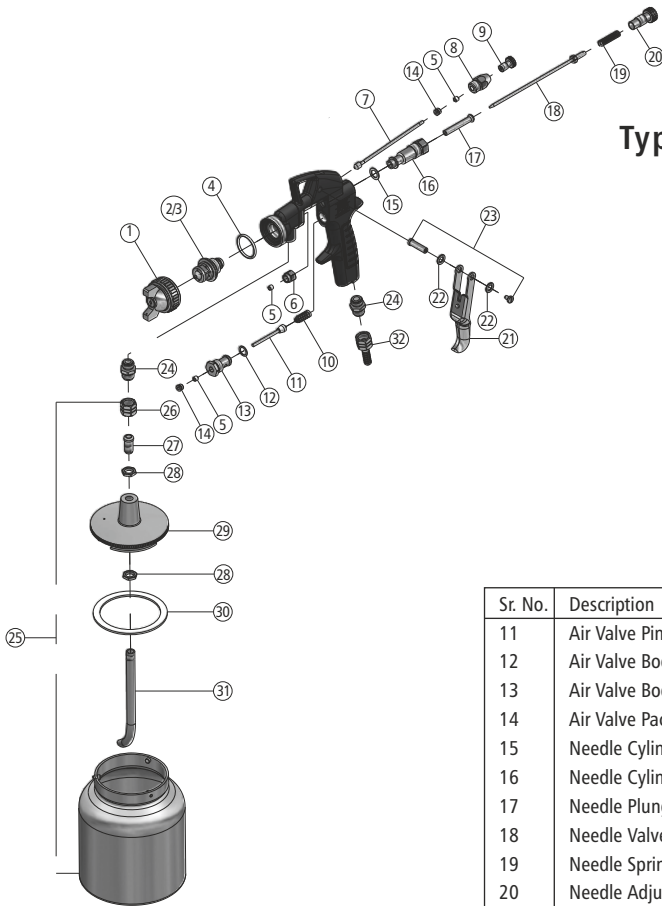
2. Remove trigger assembly (21) by unscrewing washer & screw (22&23)
3. Remove needle packing nut (6) and packing (5). Replace any worn or damaged parts and re-assemble, reversing steps 1 to 3. Adjust packing so that the movement of the needle is not restricted.

### **Air Valve Pin (11) Packing (5)**

Remove needle valve as 1 above.

4. Remove air valve body (13), pull trigger (21) unscrew air valve packing box washer (14) and remove air valve spring (10), air valve pin (11), and packing (5). Replace any worn or damaged parts and re-assemble reversing steps 1 and 4.

## Spare Parts for Type - P-70 Spray Gun



### Type P-70 Spray Gun

#### List of Spare Parts Type P-70

| Sr. No. | Description                                                   | Part No.       |
|---------|---------------------------------------------------------------|----------------|
|         | <b>Spray Gun Type P - 70<br/>with Bottom Feed Cup 1 Litre</b> | <b>70000</b>   |
|         | <b>Spray Gun Type P - 70<br/>without Cup</b>                  | <b>70000WC</b> |
| 1       | Air Cap Assembly @                                            | 70001A         |
| 2       | M. S. Nozzle @                                                | 70004          |
| 3       | S. S. Nozzle @                                                | 70005          |
| 4       | Nozzle Gasket                                                 | 70006          |
| 5       | Packing Bush Set                                              | 70008          |
| 6       | Needle Packing Nut                                            | 70009          |
| 7       | Spreader Control Spindle                                      | 70012          |
| 8       | Spreader Control Knob Guide                                   | 70014          |
| 9       | Control Knob                                                  | 70015          |
| 10      | Air Valve Spring                                              | 70016          |

| Sr. No. | Description                    | Part No. |
|---------|--------------------------------|----------|
| 11      | Air Valve Pin                  | 70017    |
| 12      | Air Valve Body Packing Bush    | 70018    |
| 13      | Air Valve Body                 | 70019    |
| 14      | Air Valve Packing Box Washer   | 70020    |
| 15      | Needle Cylinder Packing Washer | 70021    |
| 16      | Needle Cylinder                | 70022    |
| 17      | Needle Plunger                 | 70022A   |
| 18      | Needle Valve                   | 70023    |
| 19      | Needle Spring                  | 70024    |
| 20      | Needle Adjusting Screw         | 70025    |
| 21      | Trigger Assembly               | 70026    |
| 22      | Trigger Washer                 | 70027    |
| 23      | Trigger Screw                  | 70028    |
| 24      | Air Intake Connector           | 70029    |
| 25      | Suction Feed Cup 1 Litre       | 70030    |
| 26      | Coupling Nut                   | 70030A   |
| 27      | Connection Nipple              | 70030B   |
| 28      | Connection Nut                 | 70030C   |
| 29      | Cup Lid                        | 70030D   |
| 30      | Packing Washer                 | 70030E   |
| 31      | Fluid Tube                     | 70030F   |
| 32      | Hose Coupling                  | 70032    |
| 33      | M.S Repair Kit @#*             | 70080M   |
| 34      | S.S Repair Kit @#*             | 70080S   |

# Accessories

\* This set contains an air cap, nozzle, needle, air valve pin and packing.

@ Please mention nozzle size while ordering spare parts.

# Service Checks

## Troubleshooting

| Fault                                                                                                         | Cause                                                                                                                                                                                        | Remedy                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Not spraying</b>                                                                                           | <ol style="list-style-type: none"><li>1. No pressure at the gun</li><li>2. Needle adjusting Screw not properly adjusted.</li></ol>                                                           | <ol style="list-style-type: none"><li>1. Check air supply</li><li>2. Adjust</li></ol>                                                    |
| <b>Uneven spray</b><br>       | <ol style="list-style-type: none"><li>1. Material build up on air cap/nozzle</li><li>2. Coating material flow or viscosity incorrect</li></ol>                                               | <ol style="list-style-type: none"><li>1. Clean air cap / nozzle</li><li>2. Adjust needle adjustment screw, or Reduce viscosity</li></ol> |
| <b>Intermittent spray</b><br> | <ol style="list-style-type: none"><li>1. Insufficient material in cup</li><li>2. Gun material passage blocked</li><li>3. Loose or damaged nozzle</li><li>4. Needle sealing damage.</li></ol> | <ol style="list-style-type: none"><li>1. Fill cup</li><li>2. Clean</li><li>3. Tighten or replace</li><li>4. Replace</li></ol>            |
| <b>Fluid leakage from packing nut</b>                                                                         | <ol style="list-style-type: none"><li>1. Needle packing damage</li></ol>                                                                                                                     | <ol style="list-style-type: none"><li>1. Replace packing.</li></ol>                                                                      |
| <b>Dripping from the nozzle</b>                                                                               | <ol style="list-style-type: none"><li>1. Damaged nozzle or needle</li><li>2. Foreign substances between nozzle and needle prevent sealing</li></ol>                                          | <ol style="list-style-type: none"><li>1. Replace</li><li>2. Clean needle and nozzle in thinner or Replace.</li></ol>                     |

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